

## TAMIL NADU STATE BOARD - STANDARD 10 MATHEMATICS

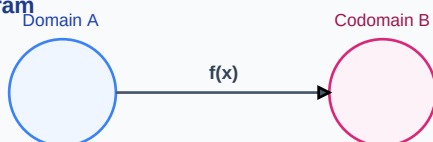
# CHAPTER 6: TRIGONOMETRY

## PREMIUM ACADEMIC STUDY SUITE &amp; EXAM GUIDE

|               |                                             |
|---------------|---------------------------------------------|
| Board:        | Tamil Nadu State Board (TNSB)               |
| Syllabus:     | Samacheer Kalvi Textbook Curriculum aligned |
| Standard:     | Class 10 English Medium                     |
| Subject:      | Mathematics                                 |
| Chapter Name: | Trigonometry                                |

**Learning Objectives:**

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

**Function Mapping Diagram**

## Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

| Term                     | Official Definition & Context                                                                                |
|--------------------------|--------------------------------------------------------------------------------------------------------------|
| Trigonometric Ratios     | Ratios of side lengths of a right-angled triangle, representing angles relative to circular projections.     |
| Trigonometric Identities | Equations involving trigonometric functions that are true for all values of the occurring variables.         |
| Angle of Elevation       | The angle formed by the line of sight and the horizontal plane for an object above the observer's eye level. |
| Angle of Depression      | The angle formed by the line of sight and the horizontal plane for an object below the observer's eye level. |

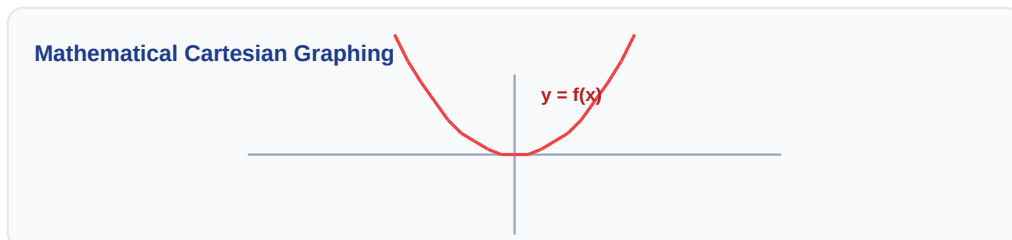
**Syllabus Exam Tip:** Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram:  $A \cap B$



## Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



### Essential Formulas, Laws & Identities:

- Fundamental Identity 1:  $\sin^2(x) + \cos^2(x) = 1$
- Fundamental Identity 2:  $1 + \tan^2(x) = \sec^2(x)$
- Fundamental Identity 3:  $1 + \cot^2(x) = \operatorname{cosec}^2(x)$
- Complementary Ratios:  $\sin(90 - x) = \cos(x)$  |  $\tan(90 - x) = \cot(x)$
- Ratios values:  $\sin(30)=0.5$ ,  $\sin(45)=1/\sqrt{2}$ ,  $\sin(60)=\sqrt{3}/2$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

## Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

**Example 1: Prove that  $[\sin(x) / (1 + \cos(x))] + [(1 + \cos(x)) / \sin(x)] = 2 \operatorname{cosec}(x)$ .**

**Step-by-step Solution:**

Cross-multiplying LHS:  $[\sin^2(x) + (1 + \cos(x))^2] / [\sin(x)(1 + \cos(x))]$ .

Expand:  $[\sin^2(x) + 1 + 2\cos(x) + \cos^2(x)] / [\sin(x)(1 + \cos(x))]$ .

Since  $\sin^2(x) + \cos^2(x) = 1$ , we get:  $[2 + 2\cos(x)] / [\sin(x)(1 + \cos(x))] = 2(1 + \cos(x)) / [\sin(x)(1 + \cos(x))] = 2 / \sin(x) = 2 \operatorname{cosec}(x)$ . Proved.

### Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

## HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

**HOTS Challenge 1: A tower stands vertically on the ground. From a point 30m away from its foot, the angle of elevation of its top is 30 degrees. Find the height of the tower.**

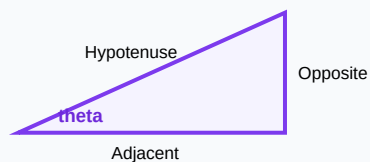
**Analytical Solution Strategy:**

Let height be  $h$ . We have adjacent side = 30m, and angle = 30 degrees.

Using tan ratio:  $\tan(30) = \text{opposite} / \text{adjacent} = h / 30$ .

Since  $\tan(30) = 1/\sqrt{3}$ , we get:  $1/\sqrt{3} = h / 30 \Rightarrow h = 30 / \sqrt{3} = 10 * \sqrt{3}$  meters (approx 17.32 m).

### Right-Angled Triangle Geometry



## Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

**Q1 (1-Mark):** Evaluate:  $\sin(30) * \cos(60) + \cos(30) * \sin(60)$ .

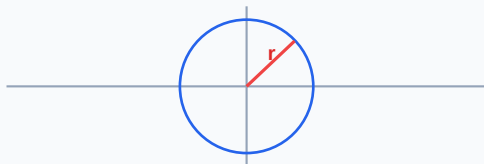
**Q2 (2-Mark):** Prove that:  $\sec(x) * (1 - \sin(x)) * (\sec(x) + \tan(x)) = 1$ .

**Q3 (2-Mark):** If  $\tan(A) = 3/4$ , find  $\sin(A)$  and  $\cos(A)$ .

**Q4 (5-Mark):** From the top of a 50m building, the angle of depression of a car is 45 degrees. Find the distance of the car from the building base.

**Q5 (5-Mark):** Show that  $\cos(90 - A) / \sin(90 - A) = \tan(A)$  is FALSE.

Circle with Radius Vector on Plane



## Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

### Application Case: Structural Surveying

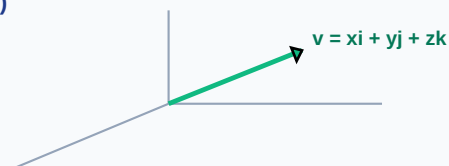
Civil engineers use theodolites to measure angles of elevation, calculating tower and bridge heights easily.

### Application Case: Satellite Telecommunications

Angles of orbit tracking are mapped using trigonometric functions to ensure perfect transmission pathways.

**Cross-Curricular Link:** Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

#### 3D Vector Projections (x, y, z)



## Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

### Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla,  $\text{cm}^3$ , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

### Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

**CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!**

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.  
Good luck!

#### Probability Tree Outcome Diagram

